



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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<https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone>



Outline

- Executive Summary
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- Methodology
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Executive Summary

- Methodologies
 - Collected data via API and web scraping
 - Cleaned data and engineered features
 - Performed exploratory data analysis (EDA) using visualization and SQL
 - Built interactive visual analytics using Folium
- Results
 - Identified key factors influencing successful landings
 - Developed classification models to predict landing success

Introduction

- Project background and context
 - SpaceX advertises Falcon 9 rocket launches on its website at a cost of \$62 million, while other providers charge upwards of \$165 million each.
 - SpaceX's Falcon 9 launches like regular rockets, but much of the savings comes from the fact that SpaceX can reuse the first stage.
 - Therefore, if we can determine whether the first stage will land, we can estimate the cost of a launch.
- Problems you want to find answers
 - What factors influence the successful landing of a Falcon 9 first stage?
 - Can we develop a machine learning model to accurately predict the landing outcome?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Data collected from the SpaceX API
 - Performed web scraping to collect Falcon 9 historical launch records from Wikipedia
- Perform data wrangling
 - Converted raw data into a cleaner and more structured format that is ready for analysis
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Build, tune and evaluate classification models

Data Collection

- From the SpaceX API
- Web scraping historical launch records from Wikipedia

Data Collection - SpaceX API

- Data collection methodology:
 - We start requesting rocket launch data from SpaceX API, using the `requests.get` method.
 - Then we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`
- GitHub URL:
 - <https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-1-Introduction/O1-Data-Collection-with-API.ipynb>

Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json'
```

We should see that the request was successful with the 200 status response code

```
response=requests.get(static_json_url)
```

```
response.status_code
```

```
200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
# Use json_normalize method to convert the json result into a dataframe  
data=pd.json_normalize(data = response.json())
```

Using the dataframe `data` print the first 5 rows

Data Collection - Scraping

- Data collection methodology:
 - Perform an HTTP GET method to request the Falcon9 Launch HTML page.
 - Create a BeautifulSoup object from the HTML response.
 - Extract all column/variable names from the HTML table header
 - Extract all column/variable names from the HTML table header
 - Create a data frame by parsing the launch HTML tables
- GitHub URL:
 - <https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-1-Introduction/O2-Data-Collection-with-Web scraping.ipynb>

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

```
# use requests.get() method with the provided static_url and headers
# assign the response to a object
response = requests.get(url=static_url, headers=headers)
response.status_code
```

200

Create a BeautifulSoup object from the HTML response

```
soup = BeautifulSoup(response.text, "html.parser")
```

Print the page title to verify if the BeautifulSoup object was created properly

```
soup.title
```

```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about BeautifulSoup, please check the external reference link towards the end of this lab

```
# Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`
html_tables = soup.find_all("table")
```

Starting from the third table is our target table contains the actual launch records.

```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
```

Data Wrangling

- How data were processed:
 - Cleanup missing values
 - Calculate the number of launches on each site
 - Calculate the number and occurrence of each orbit
 - Calculate the number and occurrence of mission outcome of the orbits
 - Create a landing outcome label from Outcome column
- GitHub URL:
 - <https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-1-Introduction/O3-Data-Wrangling.ipynb>

EDA with Data Visualization

- Charts plotted and why:
 - Scatter plot to visualize the relationship between Flight Number and Launch Site
 - Scatter plot to visualize the relationship between Payload Mass and Launch Site
 - Bar chart to visualize the success rate for each orbit type
 - Scatter plot to visualize the relationship between FlightNumber and Orbit type
 - Line chart to Visualize the launch success yearly trend
- GitHub URL:
 - [https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-2-Exploratory-Data-Analysis-\(EDA\)/05-EDA-with-Visualization.ipynb](https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-2-Exploratory-Data-Analysis-(EDA)/05-EDA-with-Visualization.ipynb)

EDA with SQL

- SQL queries performed:

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first succesful landing outcome in ground pad was acheived.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

- GitHub URL:

- [https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-2-Exploratory-Data-Analysis-\(EDA\)/04-EDA-with-SQL.ipynb](https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-2-Exploratory-Data-Analysis-(EDA)/04-EDA-with-SQL.ipynb)

Build an Interactive Map with Folium

- Map created and added to the folium map:
 - Used folium.Circle to mark all launch sites on a map
 - Used marker_cluster and folium.Marker to mark the success/failed launches for each site on the map
 - Used folium.PolyLine to draw a line from the launch site to the closest coastline, railway, highway
- GitHub URL:
 - <https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-3-Interactive-Visual-Analytics-and-Dashboards/06-Launch-Site-Location-Visual-Analytics-with-Folium.ipynb>

Build a Dashboard with Plotly Dash

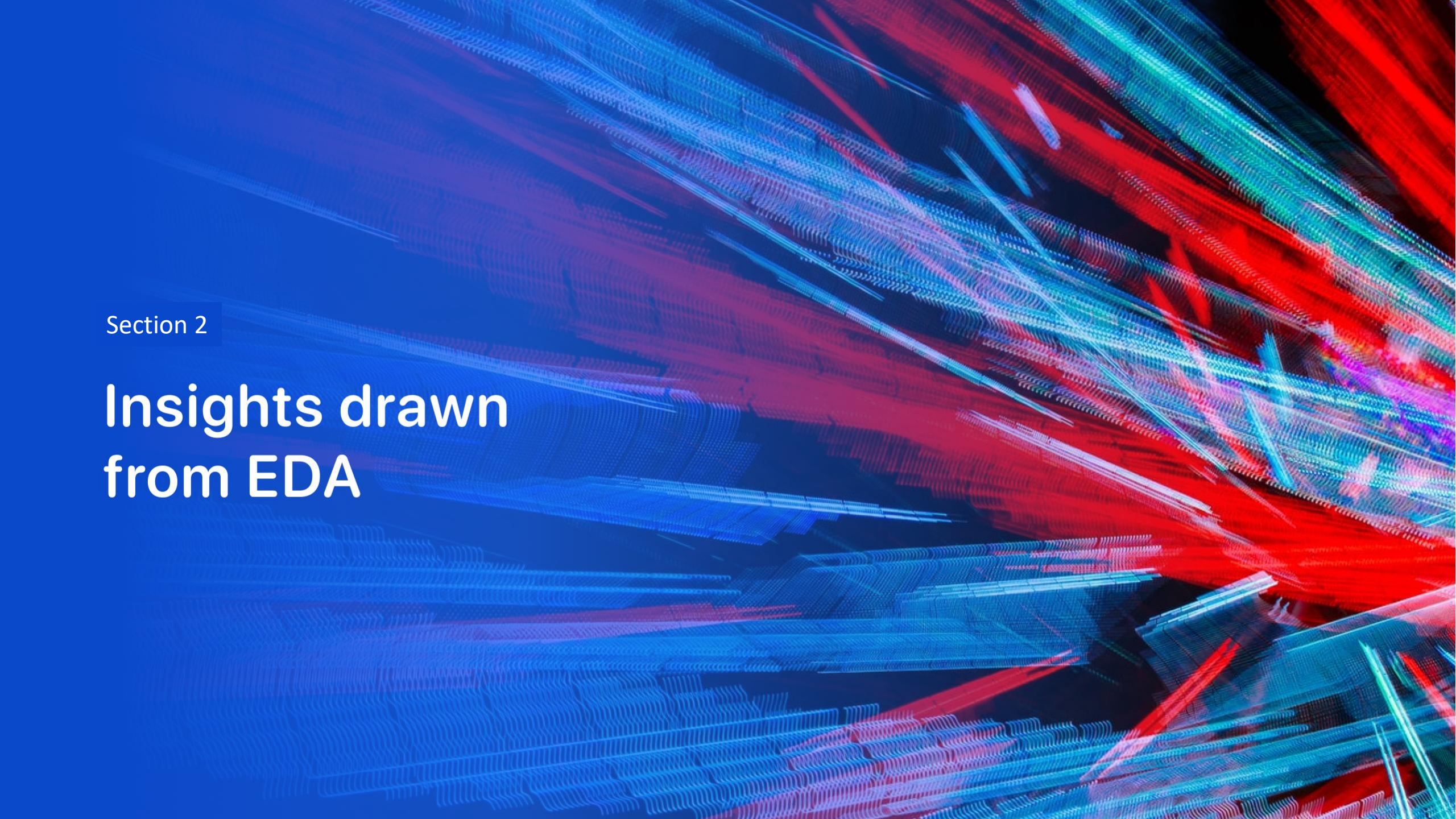
- Plots/graphs and interactions you have added to a dashboard:
 - Pie Charts to show the total launches by site.
 - Scatter plots to show the relationship between payload mass and launch success
- GitHub URL:
 - <https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-3-Interactive-Visual-Analytics-and-Dashboards/spacex-dash-app.py>

Predictive Analysis (Classification)

- Explored different classification models:
 - Logistic Regression
 - Decision Tree
 - K Nearest Neighbors
 - Support Vector Classifier
- Model Tuning:
 - Used GridSearchCV to find the best hyperparameters for the machine learning models
- Model Evaluation:
 - Cross-Validation Techniques
 - Calculated the accuracy of the models and identified which one performs the best
- GitHub URL:
 - <https://github.com/giovannibianchinidebarros/IBM-Applied-Data-Science-Capstone/blob/main/Module-4-Predictive-Analysis/O7-Machine-Learning-Prediction.ipynb>

Results

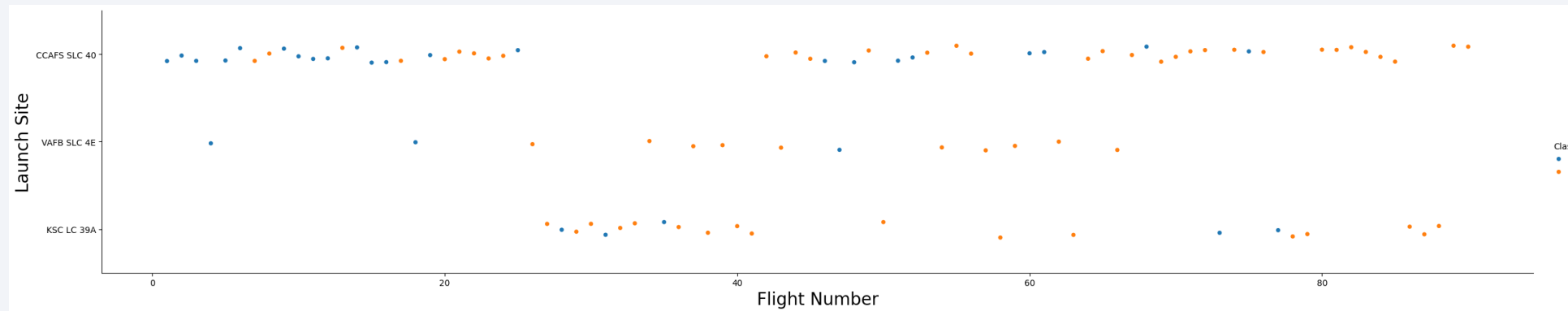
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

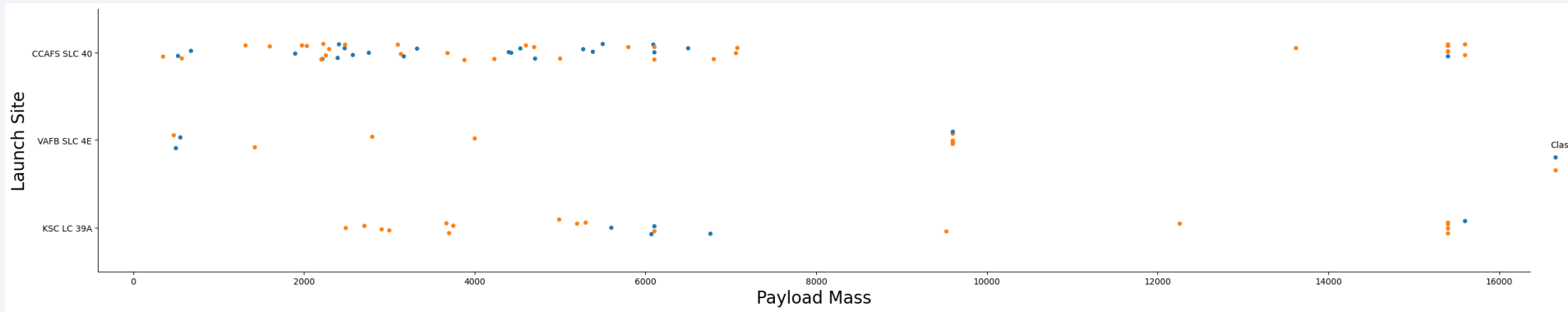
Insights drawn from EDA

Flight Number vs. Launch Site



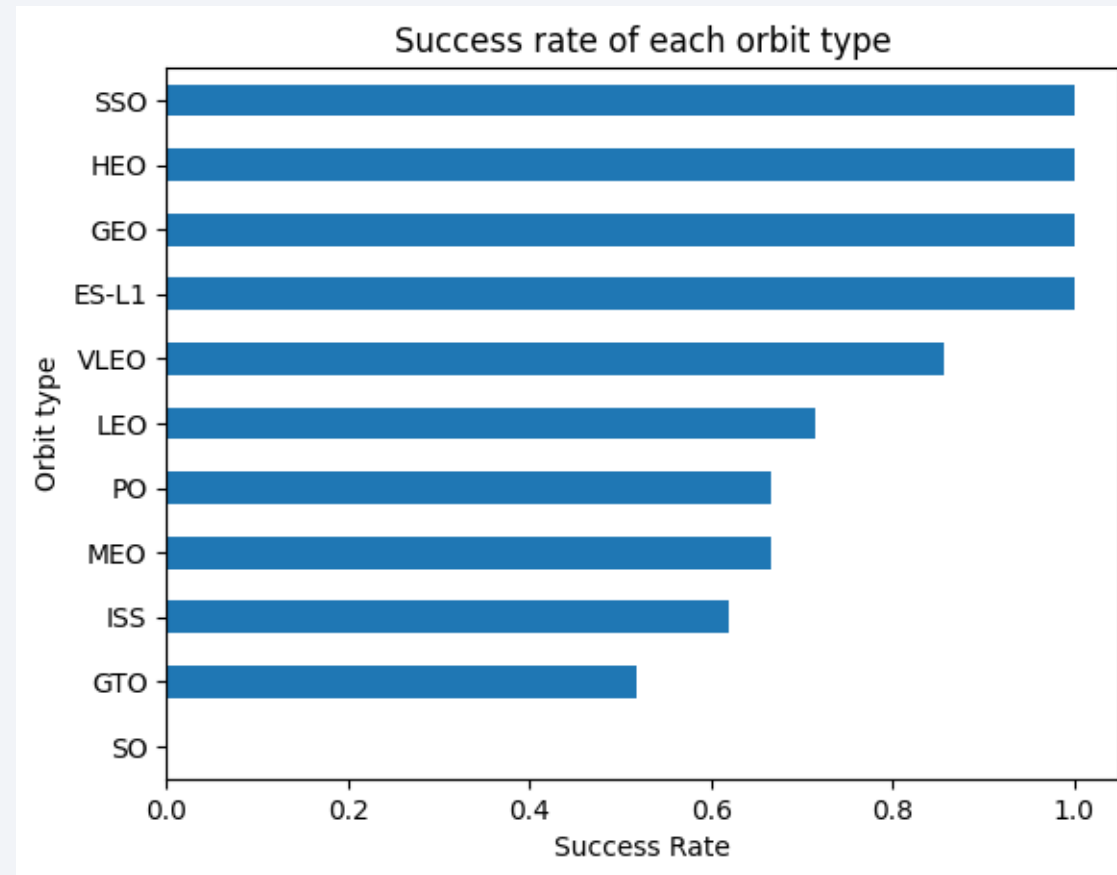
- All sites feature a combination of successful and failed landings.

Payload vs. Launch Site

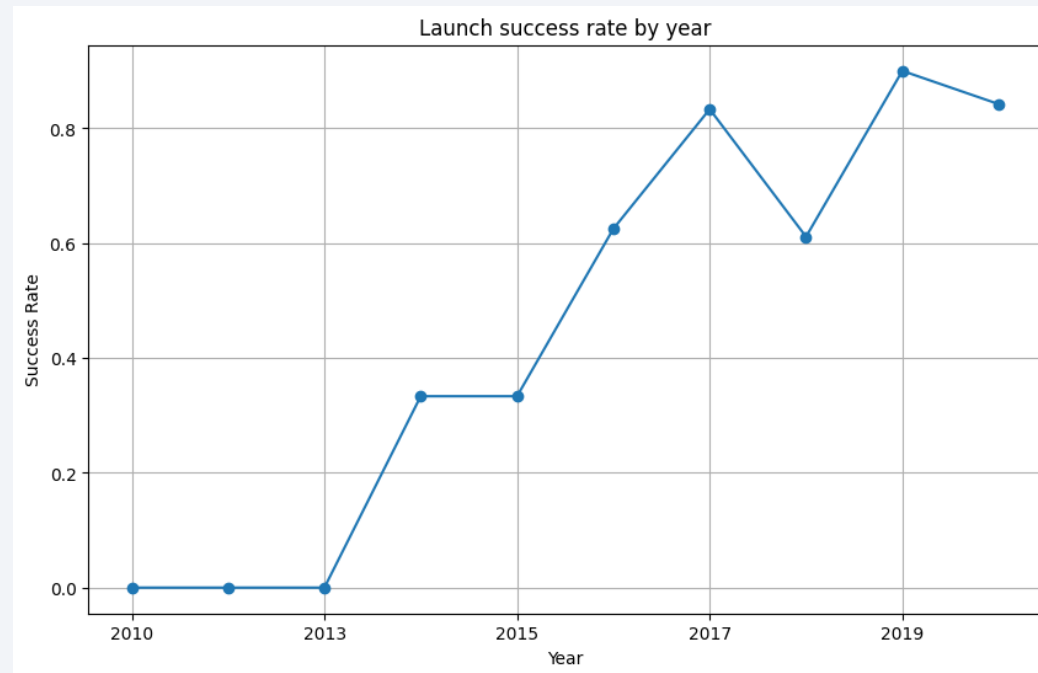


- For the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000)

Success Rate vs. Orbit Type



Launch Success Yearly Trend



- We can observe that the success rate since 2013 kept increasing till 2020

All Launch Site Names

CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Conclusions

- Success rate since 2013 kept increasing till 2020
- SSO, LEO, GEO and ES-L1 Orbits had the best success rates.
- Decision Tree classification had the best model score

Thank you!

